

Fall 2025 Extreme temperatures and public health - Executive Summary

Purpose: This project quantifies how short periods of hot and cold weather affect mortality across 10 UK regions. We used data for deaths, published by the UK government, covering the period of 1981 to 2022. However, 2020 onwards is excluded due to COVID related distortion. The health data is combined with weather data covering the same period. Our project aimed to inform surge planning and longer term climate–health policy.

Approach: We constructed population adjusted weekly excess mortality, removed long run and seasonal patterns with LOESS smoothing and linked the de-seasonalised series to weekly aggregated weather features. Two modelling tracks were used: (1) a simple quadratic fit between weekly temperature and excess deaths, and (2) a per region CNN using various combinations of temperature, humidity, snow, wind and precipitation. Separately, an epidemiological Distributed Lag Non linear Model (DLNM) was fitted on daily data to explain (not predict) the temperature–mortality relationship and identify risk thresholds.

Key findings:

- After removing seasonality caused by e.g. winter flu, we continue to see excess mortality at the highest and lowest temperatures. However, although this correlation is present, it is weak. The short term excess deaths are dominated by variability that is significantly larger than weather impacts. The CNN only marginally beats a zero baseline and the simple quadratic performs almost as well.
- Cold risk is universal across all regions analysed. Mortality risk becomes statistically significant around 9 °C and below.
- Heat risk is regional. Southern and Midlands regions show significant heat effects. For example, London's minimum mortality temperature is 18 °C and its heat threshold is 24 °C, while some other regions show no statistically significant heat effects.
- Explanatory power is strong at daily resolution. DLNM fits explain roughly 53–82% of variance depending on region. For example, London has an R^2 value of 0.82, providing robust interpretable temperature–risk curves for planning and thresholds.

Implications for stakeholders and next steps:

- Periods of extreme heat and cold have a tangible effect of public health. They cause an increase in excess deaths. However, this effect is smaller than the random fluctuations (noise) in the weekly deaths data.
- Our models are therefore unlikely to be useful in predicting the need for surge planning, as was initially envisioned. However, although the number of excess deaths is small, it is significant. The several dozen excess deaths predicted during a heatwave month in some regions may justify public health campaigns during extreme weather.
- An interesting but incomplete look at time lag effects near the end of the project appears to show promising results and may improve the predictive power of our model.
- Our models are likely suited to predicting longer term effects e.g. how a long period of warm weather or climate change will affect public health in the long term.